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ORIGINAL INVESTIGATION

Single Session Intermittent Heat Exposure With More Frequent and Shorter Cooling Breaks Facilitates Greater Training Intensity and Elicits Physiological Responses Comparable to Continuous Heat Exposure

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Purpose: To investigate the influence of shorter, more frequent rest breaks with per-cooling as an alternative heat-acclimation session on physiological, perceptual, and self-paced maximal cycling performance, compared with continuous heat exposure. ***Methods:*** Thirteen participants completed 1 continuous and 3 intermittent-heat-exposure (IHE) maximal self-paced cycling protocols in a random order in heat (36 °C, 80% relative humidity): 1 × 60-minute exercise (CON), 3 × 20-minute exercise with 7.5-minute rest between sets (IHE-20), 4 × 15-minute exercise with 5-minute rest between sets (IHE-15), and 6 × 10-minute exercise with 3-minute rest between sets (IHE-10). Mixed-method per-cooling (crushed-ice ingestion and cooling vest) was applied during rest periods of all IHE protocols. ***Results:*** Total distance completed was greater in IHE-10, IHE-15, and IHE-20 than in CON (+11%, +9%, and +8%, respectively), with no difference observed between IHE protocols. Total time spent above 38.5 °C core temperature was longer in CON compared with IHE-15 and IHE-20 (+62% and +78%, respectively) but similar to IHE-10 (+5%). Furthermore, a longer time above 38.5 °C core temperature occurred in IHE-10 versus IHE-15 and IHE-20 (+54% and +69%, respectively). Sweat loss did not differ between conditions. ***Conclusion:*** IHE with per-cooling may be a viable alternative heat-acclimation protocol in situations where training quality takes precedence over thermal stimulus or when both factors hold equal priority.

Keywords: heat acclimation, self-paced, cycling

Heat acclimation (HA) is critical to mitigate the adverse effects of heat on exercise performance.[1] Traditional HA protocols involve exercise bouts in hot environments (>30 °C) lasting 60 to 120 minutes over 10 to 14 days.[2,3] To elicit heat adaptations, including reductions in exercising and resting heart rate (HR), skin (T_{sk}) and core temperature (T_c), expansion of plasma volume, increased sweat rate, and improvements in thermal comfort (Th_c) and sensation (Th_s), various methodologies have been used.[2] These methodologies encompass controlled hyperthermia (eg, maintaining T_c at 38.5 °C throughout exercise), constant work rate (eg, exercising at 60% maximal aerobic capacity), controlled intensity (eg, exercising at 65% maximal HR), and/or passive heating (eg, water immersion or sauna bathing). While these methods yield physiological adaptations to heat, they also result in greater internal training load.[4] This can potentially compromise training quality and lead to overreaching. Therefore, there is a need for HA protocols that address situations where training quality is crucial, yet the heat stimulus is necessary.

Another method of HA that may maintain training quality without compromising heat stimulus is self-paced exercise. This approach allows athletes to freely adjust their pacing to attain or maintain intensity around a prescribed perception.[1,2,5] Furthermore, acute cooling such as precooling and per-cooling (PerC) may assist in preserving exercise quality during HA sessions, despite appearing contradictory to the aims of HA training (ie, increase T_c and T_{sk}, and induce sweating).[1] However, application of cooling can blunt increases in T_c and T_{sk}; improve Th_c, Th_s, and effort; and thus improve/maintain training and exercise performance.[6] Choo et al[7] investigated the use of precooling (30 min of ~22 °C cold-water immersion) prior to 10 HA sessions (60-min cycling at a rating of perceived exertion [RPE] of 15[8]) in heat (~35 °C ~53% relative humidity [RH]). Their findings showed maintenance of a greater mean power output over the 10-day period with precooling (+2.9%) compared to no precooling (–2.6%). Similarly, Naito et al[9] found that using PerC (via crushed-ice ingestion) during 5 HA sessions (80-min intermittent repeated sprint protocol in 36.5 °C, 50% RH) resulted in greater total work observed on day 1 (approximately +4%) and day 5 (approximately +3%) of the HA sessions compared with no PerC. This may be attributed to lower end-exercise rectal temperature and T_{sk} in the PerC condition compared with no PerC on day 1 (approximately –0.4 and –1.0 °C, respectively) and day 5 (approximately –0.4 and –0.4 °C, respectively). These findings suggest that incorporating cooling during HA training sessions facilitates greater training intensity, presenting a viable training alternative for a HA program.

Another potential method of reducing the internal training load and mitigating athlete fatigue during HA involves using an intermittent heat exposure (IHE) protocol. This approach exposes athletes to heat during exercise intervals, followed by passive rest in thermoneutral conditions. To our knowledge, the use of IHE as a means of HA has not been previously explored. Potentially IHE could maintain and/or improve training quality by implementing cooling strategies (ie, PerC) during rest breaks, thus benefiting from the ergogenic effects of PerC compared with no PerC.[10]

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However, before implementing IHE with PerC into a full HA protocol (chronic effects), we first need to better understand the performance, physiological, and perceptual responses to different rest breaks with PerC at the level of a single session (acute effects).

Therefore, the aim of this study was to investigate the acute physiological, perceptual, and performance responses to shorter, more frequent rest breaks with PerC as an alternative HA session compared to a traditional continuous HA session without cooling in nonheat acclimatized endurance athletes. To understand the effect of different IHE set structures (ie, short, frequent breaks vs longer, less frequent breaks) on exercise responses, we compared 3 structures matched for total exercise and rest time. We hypothesized that training intensity and thermal perception would improve in the IHE sessions with shorter, more frequent cooling breaks while producing acute physiological responses comparable to those observed in a traditional continuous heat exposure.

Methods

Participants

Thirteen male cyclists and/or triathletes (mean [SD]: age = 34.0 [11.1] y; body mass [BM] = 78.08 [9.37] kg; height =177.79 [6.30] cm; sum of 8 skinfolds = 77.27 [26.57] mm; body surface area = 1.96 [0.14] m^2;[11] training distance = 241 [120] km/wk; training frequency = 5 [2] sessions/wk), corresponding with an athlete classification of 2 or 3 (trained or highly trained[12]) provided written informed consent. Participants had not undertaken a HA protocol, trained in environments exceeding 30 °C, or resided in a summer (warm) climate in the last month (ie, nonheat acclimatized). The study was approved by the Curtin University human research ethics committee (HRE2021-0398).

Study Overview

Testing occurred during the cooler months in Western Australia (March–October) to control for exposure to high environmental temperatures and RH, which may induce heat adaptations and influence the cooling intervention. Participants attended a familiarization session followed by 4 experimental sessions, separated by 7 (2) days. These sessions were completed in a randomized order (http://www.randomization.com/, January 2022). All experimental sessions comprised self-paced cycling exercise in a heat chamber (35.8 °C, 77.7% RH; ~48 °C apparent temperature, no fan and radiant heat[13]) for a total of 60 minutes with 15-minute rest in a thermoneutral environment. During rest, mixed-method (crushed-ice plus cooling vest) PerC was applied. The 4 experimental sessions had different set structures: 1 × 60-minute exercise (CON), 3 × 20-minute exercise with 7.5-minute rest between sets (IHE-20), 4 × 15-minute exercise with 5-minute rest (IHE-15), and 6 × 10-minute exercise with 3-minute rest (IHE-10). Participants refrained from vigorous exercise, alcohol (within 24 h), and caffeine (within 6 h) prior to each experimental session. Additionally, participants were instructed to keep a food diary and replicate their dietary intake for the 24 hours prior to each experimental session to ensure nutritional consistency. To ensure adequate hydration prior to each experimental session, participants were instructed to drink 6 $mL \cdot kg^{-1}$ BM of water every 2.5 hour on the preceding day.

Familiarization Session

During the familiarization session, participants' anthropometric measures (height, BM, and sum of 8 skinfolds) were obtained. Standing height and BM were recorded using a calibrated stadiometer (Holtain Limited) and electronic scales (Seca 360° Wireless, Seca), respectively. Skinfold thickness was assessed using International Society for Advancement of Kinanthropometry standardized methodology at 8 sites (triceps, subscapular, biceps, iliac crest, supraspinale, abdominal, front thigh, and medial calf) using calibrated Harpenden skinfold calipers (Baty International) by an International Society for Advancement of Kinanthropometry accredited Level 1 anthropometrist (intratechnical error measurement = 1.1%). Subsequently, participants completed the entire IHE-15 protocol on a cycling ergometer (BikeErg, Concept 2) to become familiar with the demands of experimental sessions.

Experimental Session

Eight hours prior to each experimental session, participants ingested a core body temperature (T_b) pill (E-Celsius Performance, Bodycap Medical) to measure T_c. Upon arrival at the laboratory, hydration status was determined by urine specific gravity (USG; portable refractometer, RS PRO, RS Components). Participants with a USG reading of >1.021 μg were instructed to ingest water and retested every 10 minutes until USG readings were ≤1.021 μg. Nude BM was measured preexperimental and postexperimental session on a set of calibrated electronic scales (Model CW11, OHAUS Corporation) to calculate sweat loss (prenude BM + fluid ingested and postnude BM + fluid ingested). Surface T_{sk} was measured by placing dermal temperature sensors (DS1922L; iButton, Maxim Integrated Products) on the mesosternale, midforearm, midquadriceps, and medial calf to calculate mean T_{sk}.[14] A HR monitor (A300 Polar, Electro) was also placed on the participants' chest to measure HR.

Participants entered the heat chamber and completed a 10-minute cycling warm-up at a perceptually regulated exercise intensity (RPE of 13, 15, and 17 on the 6–20 Borg scale for 5, 3, and 2 min, respectively[8]) on the same cycling ergometer as the familiarization session. This was followed by completing the first exercise interval at the participant's highest sustainable intensity. Participants were blinded to their completed distance but received continuous feedback on time remaining. After each exercise interval, participants exited the heat chamber and moved to a thermoneutral environment (23.7 °C, 53.9% RH; ~25 °C apparent temperature) to passively rest and receive the PerC intervention. During the rest period, participants sat wearing an activated cooling vest (as per manufacturer's instructions, Arctic Heat). At the same time, crushed-ice (using an ice shaver, Avalanche Model IS6800, Sunbeam) was ingested to achieve a session total of 3.5 $g \cdot kg^{-1}$ BM, which represents 15 minutes of previously utilized ingestion rate (IHE-10: 0.7 $g \cdot kg^{-1}$ BM per rest break, IHE-15: 1.17 $g \cdot kg^{-1}$ BM, and IHE-20: 1.75 $g \cdot kg^{-1}$ BM).[15] No cooling occurred during or after the CON trial. Upon completion of the rest interval, participants reentered the heat chamber and repeated this process for all exercise/rest intervals. Participants had access to ad libitum water consumption throughout all sessions. Water was retrieved from a water fountain (~16 °C), held in a noninsulated water bottle which entered the chamber when participants commenced the warm-up, and remained in the heat chamber until a refill was necessary. After refilling, the water bottle was returned and remained in the heat chamber. HR, T_c, and T_{sk} data were collected every 2 minutes during the warm-up, and every 5 minutes during exercise intervals, and prerest and postrest intervals. T_c was monitored continuously (15-s sample rate) throughout the session to calculate time spent above 38.5 °C T_c. Th_c and Th_s were recorded (0–20 scales ranging from "very cold" to "very hot" and from "very comfortable" to

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"very uncomfortable"[16]) prewarm-up and postwarm-up, and every 5 minutes during exercise and prerest and postrest intervals. RPE (6–20 scale[8]) was obtained every 5 minutes during the exercise protocol. Exercise performance was assessed by measuring split distance every 5 minutes as well as total distance completed.

Data and Statistical Analyses

T_b was calculated using the equation: $0.65 \times T_c + 0.35 \times T_{sk}$.[17] Intraclass correlation coefficients were calculated for all variables to determine the percentage of variance explained by individual differences between participants. When intraclass correlation coefficient values exceed .5, analysts are advised to utilize statistical models that account for nonindependence in the data.[18] Given data sampled repeatedly from participants were not independent and intraclass correlation coefficients in 9 of 20 outcomes were >.5 (see Supplementary Materials [available online]), multilevel modeling was used to analyze the data. Multilevel modeling analyzed the relationship between T_c, mean T_{sk}, T_b, HR, RPE, Th_c, Th_s, and total and split distance completed for each condition, with estimates reported as unstandardized regression coefficients (b). This indicates the amount of change (in the unit of measurement, eg, beats per minute and degree Celsius) between different levels of categorical independent variables (ie, condition) or per second increase in time.[19] Additionally, linear mixed-effects regression model was used to determine differences in hydration, total fluid ingested, sweat loss, and time spent above 38.5 °C T_c during exercise intervals. Where significant interaction effects were identified, post hoc simple slopes analyses, corrected for multiple comparisons, were conducted to compare the relationship between distance and condition pairings. Key findings are summarized in the text, and full results are provided as Supplementary Materials (available online). Linear mixed-effects regression and post hoc analyses were conducted using the "lme4" and "emmeans" packages, respectively, in R with significance accepted at $P < .05$.[20,21]

Results

Hydration and Sweat Loss

USG was significantly lower in CON (1.011 [0.010] µg) compared with IHE-10 (1.016 [0.010] µg, $b = -0.01$, $P < .01$) and IHE-15 (1.015 [0.010] µg, $b = -0.004$, $P < .01$), but not in IHE-20 (1.012 [0.010] µg, $b = -0.001$, $P = .32$).

Total fluid ingestion was greater in CON (1.85 [0.57] L) compared with IHE-10 (1.49 [0.67] L, $b = 0.36$, $P < .01$), IHE-15 (1.36 [0.67] L, $b = 0.50$, $P < .01$), and IHE-20 (1.53 [0.82] L, $b = 0.32$, $P < .01$).

No significant difference in sweat loss was observed between CON (2.12 [0.53] L) and IHE-10 (2.07 [0.65] L, $b = -0.05$, $P = .39$), IHE-15 (2.07 [0.56] L, $b = -0.05$, $P = .30$), and IHE-20 (2.02 [0.60] L, $b = -0.10$, $P = .06$).

Cycling Performance

Total distance completed was 11%, 9%, and 8% greater in IHE-10 (26,765 [1300] m, $b = -2203.88$, $P < .01$), IHE-15 (26,406 [1300] m, $b = -1928.64$, $P < .01$), and IHE-20 (26,167 [1330] m, $b = -1689.95$, $P < .01$) than CON (24,232 [1858] m), respectively. No significant difference was observed between IHE protocols. There was a significant ($P < .01$) 2-way interaction between condition and time in IHE-10 ($b = 4.11$), IHE-15 ($b = 4.79$), and IHE-20 ($b = 4.73$) for split distance completed. Post hoc comparisons (Figure 1) indicated lower split distance completed over time in CON than all IHE protocols. No differences in split completion distance were identified between IHE protocols.

Thermal Response

Exercise

Accumulated time (and percentage of total session) above 38.5 °C T_c was longer in CON (30.3 [18.5] min, ~50%) compared with IHE-15 (18.7 [23.2] min, ~31%, $b = -500.28$, $P = .03$) and IHE-20 (17.0 [27.0] s, ~28%, $b = -893.23$, $P < .01$). No difference was observed between IHE-10 and CON (28.8 [24.5] min, ~48%, $b = 32.19$, $P = .89$). Longer time above 38.5 °C T_c was observed in IHE-10 compared with IHE-15 ($b = -532.47$, $P = .02$) and IHE-20 ($b = -925.43$, $P < .01$). There was a significant 2-way interaction (condition × time) in all IHE protocols for T_c (IHE-10: $b = -0.02$, $P < .01$, IHE-15: $b = -0.02$, $P < .01$, IHE-20: $b = -0.03$, $P < .01$); mean T_{sk} (IHE-10: $b = -0.02$, $P < .01$, IHE-15: $b = -0.02$, $P < .01$, IHE-20: $b = -0.02$, $P < .01$); and T_b (IHE-10: $b = -0.02$, $P < .01$, IHE-15: $b = -0.02$, $P < .01$, IHE-20: $b = -0.03$, $P < .01$) during exercise intervals. Post hoc comparisons indicated a greater increase in T_c (Figure 2A) over time in CON than all IHE protocols (IHE-10: $b = 0.02$, $t[509] = 7.14$, $P < .01$, IHE-15: $b = 0.02$, $t[509] = 6.38$, $P < .01$, IHE-20: $b = 0.03$, $t[509] = 9.31$, $P < .01$) and greater increase over time in IHE-15 compared to IHE-20 ($b = 0.01$, $t[509] = 3.55$, $P < .01$). Similarly, post hoc comparisons indicated greater mean T_{sk} (Figure 2A) over time in CON than all IHE protocols (IHE-10: $b = 0.02$, $t[584] = 4.68$, $P < .01$, IHE-15: $b = 0.02$, $t[584] = 5.07$, $P < .01$, IHE-20: $b = 0.02$, $t[584] = 4.69$, $P < .01$). No significant difference between IHE protocols was observed. Post hoc comparisons for T_b (Figure 3) indicated greater T_b over time in CON than all IHE protocols (IHE-10: $b = 0.02$, $t[509] = 7.23$, $P < .01$, IHE-15: $b = 0.003$, $t[509] = 7.06$, $P < .01$, IHE-20: $b = 0.03$, $t[509] = 9.10$, $P < .01$) and greater over time in IHE-15 compared to IHE-20 ($b = 0.01$, $t[509] = 2.60$, $P < .01$).

There was a significant 2-way interaction (condition × time) for HR in IHE-20 ($b = -0.24$, $P < .01$). Post hoc comparisons (Figure 4) indicated greater HR over time in CON than IHE-20 ($b = 0.24$, $t[580] = 2.89$, $P = .02$), and no difference compared to IHE-15 ($b = 0.14$, $t[580] = 1.75$, $P = .30$) and IHE-10 ($b = 0.18$, $t[580] = 2.15$, $P = .14$). No significant difference between IHE protocols was observed.

Rest

There was a significant 2-way interaction (condition × time) in IHE-20 for T_c ($b = -0.06$, $P < .01$); mean T_{sk} ($b = -0.13$, $P < .01$); T_b ($b = -0.10$, $P < .01$); and HR ($b = -2.65$, $P < .01$) during the rest periods. Post hoc comparisons indicated lower T_c (Figure 5; $b = 0.06$, $t[204] = 3.81$, $P < .01$), mean T_{sk} ($b = 0.14$, $t[242] = 4.07$, $P < .01$); T_b ($b = 0.10$, $t[204] = 5.34$, $P < .01$); and HR ($b = 2.65$, $t[220] = 3.28$, $P < .01$) over time in IHE-20 compared with IHE-10 during rest periods. Similarly, lower T_c ($b = 0.05$, $t[204] = 2.97$, $P = .01$) and T_b ($b = 0.05$, $t[204] = 2.80$, $P = .02$) over time were observed in IHE-20 compared with IHE-15, and lower T_b in IHE-15 compared with IHE-10 ($b = 0.04$, $t[204] = 2.70$, $P = .02$) during rest periods.

Perceptual Response

Exercise

There was a significant 2-way interaction (condition × time) for Th_c ($b = -0.07$, $P < .01$); Th_s ($b = -0.06$, $P < .01$); and RPE ($b = -0.02$,

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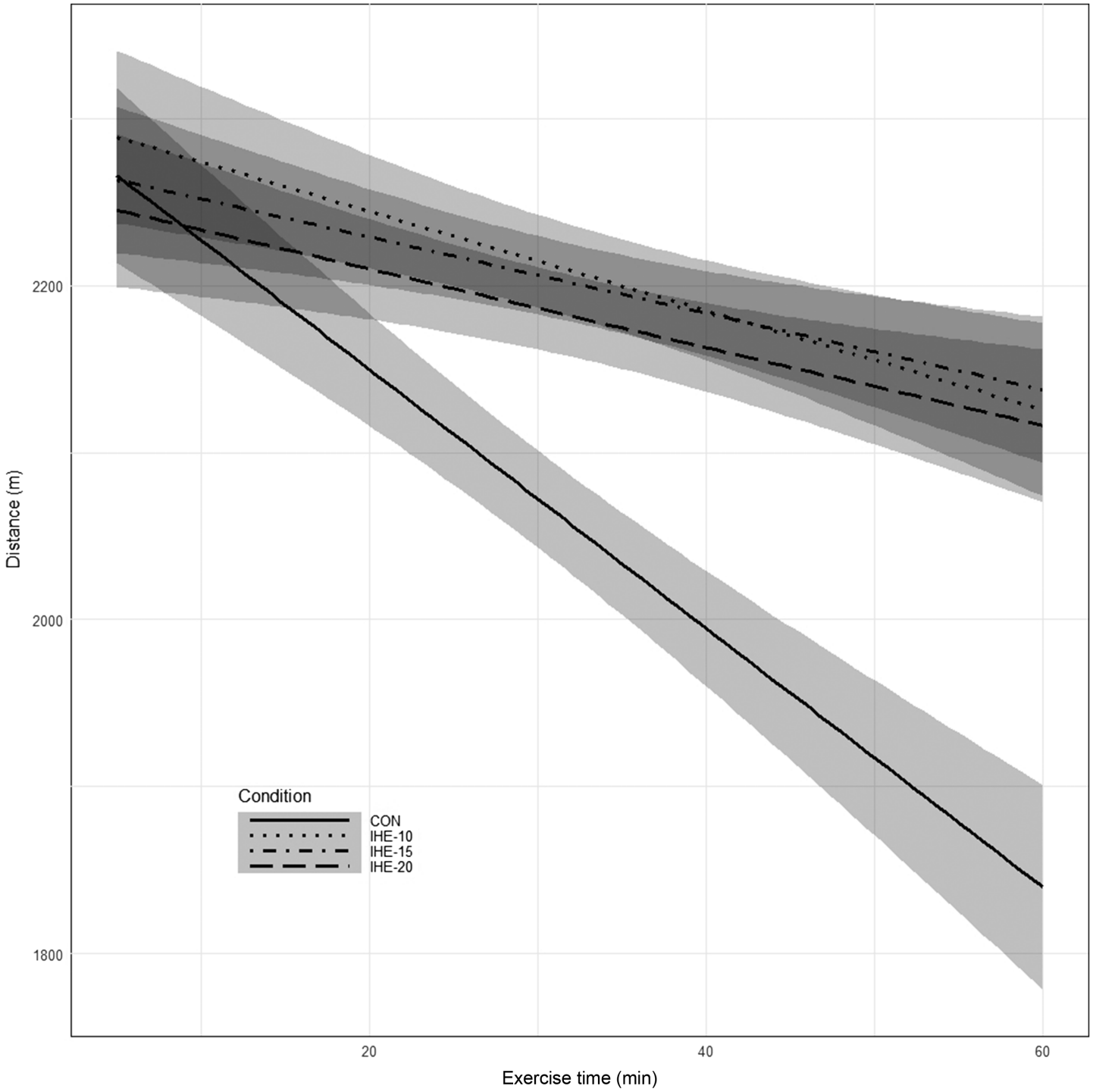

Figure 1 — Split distance completed as a function of total time cycled in CON, 6 × 10-minute (IHE-10), 4 × 15-minute (IHE-15), and 3 × 20-minute (IHE-20) IHE conditions. Note: Shaded areas indicate 95% CIs. CON indicates control; IHE, intermittent heat exposure.

$P < .01$) for IHE-20. Post hoc comparisons indicated improved Th_c (IHE-10: $b = 0.09$, $t[584] = 7.72$, $P < .01$, IHE-15: $b = 0.09$, $t[584] = 7.63$, $P < .01$, IHE-20: $b = 0.07$, $t[584] = 6.15$, $P < .01$); Th_s (IHE-10: $b = 0.07$, $t[584] = 6.89$, $P < .01$, IHE-15: $b = 0.06$, $t[584] = 6.48$, $P < .01$, IHE-20: $b = 0.06$, $t[584] = 5.86$, $P < .01$); and RPE (IHE-10: $b = 0.03$, $t[584] = 3.44$, $P < .01$, IHE-15: $b = 0.03$, $t[584] = 3.18$, $P < .01$, IHE-20: $b = 0.02$, $t[584] = 2.78$, $P < .01$) over time in all IHE protocols compared with CON. No difference between IHE protocols was observed.

Rest

There was a significant 2-way interaction (condition × time) for Th_c ($b = -0.18$, $P < .01$) and Th_s ($b = -0.18$, $P < .01$) in IHE-20. Post hoc comparisons indicated lower Th_c and Th_s (Figure 6) over time

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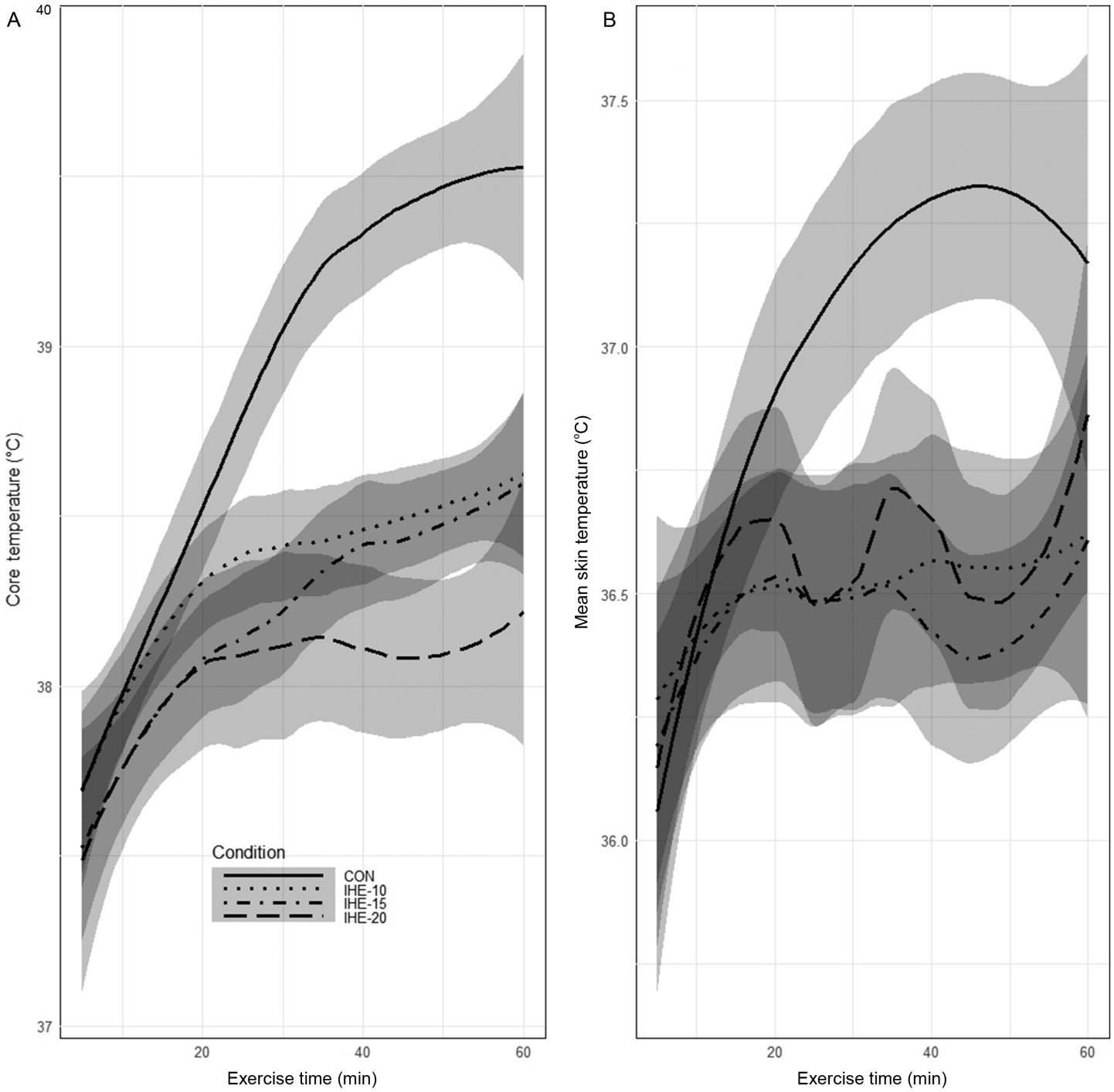

Figure 2 — (A) Core and (B) mean skin temperature as a function of total time cycled in CON, 6 × 10-minute (IHE-10), 4 × 15-minute (IHE-15), and 3 × 20-minute (IHE-20) IHE conditions. Note: Shaded areas indicate 95% CIs. CON indicates control; IHE, intermittent heat exposure.

in IHE-20 compared with IHE-10 ($b = 0.18$, $t[242] = 3.71$, $P < .01$, and $b = 0.18$, $t[241] = 3.15$, $P = .01$, respectively).

Discussion

To the best of our knowledge, this is the first study examining the effect of shorter, more frequent rest breaks with PerC as an alternative HA session compared with traditional continuous heat exposure with no cooling. One key finding was that 5-minute split distances (total and mean) were greater in all IHE protocols compared with CON, with no discernible difference between IHE protocols. Additionally, time spent above 38.5 °C T_c was not significantly different between CON and IHE-10 but was longer in CON compared with IHE-15 and IHE-20. Th_c, Th_s, and RPE were lower in all IHE protocols compared with CON, and no difference between conditions was observed for sweat loss. These results confirm our hypothesis that IHE sessions with shorter exercise durations and more frequent breaks can result in improved

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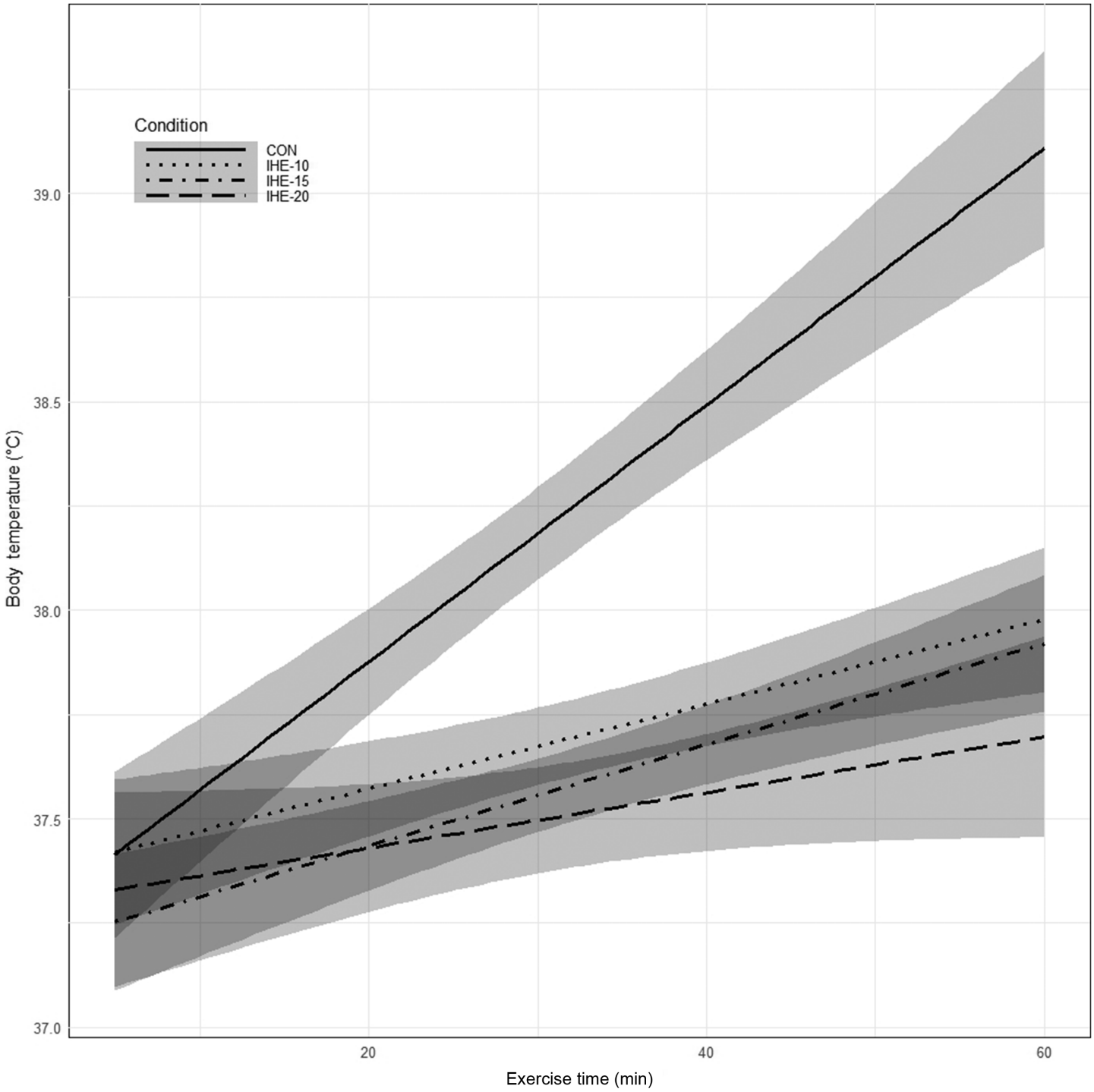

Figure 3 — Body temperature as a function of total time cycled in CON, 6 × 10-minute (IHE-10), 4 × 15-minute (IHE-15), and 3 × 20-minute (IHE-20) IHE conditions. Note: Shaded areas indicate 95% CIs. CON indicates control; IHE, intermittent heat exposure.

self-paced maximal cycling performance and thermal perception compared with continuous heat exposure. Importantly, IHE-10 with shorter exercise bout durations and more frequent breaks achieved these performance and perceptual benefits without compromising necessary acute physiological stimuli required for HA (ie, time spent above 38.5 °C T_c and sweat loss).

Enhanced performance in IHE protocols supports previous findings by Naito et al,[9] who observed significantly greater total work following PerC compared to control on day 1 (~206 vs ~198 kJ) of HA training. Performance improvements were attributed to lower T_c and T_{sk} resulting from PerC. Our study also observed lower T_b in all IHE protocols compared to CON, which may explain improved performance in heat.

T_c increased over time for all conditions but consistently remained lower during all IHE protocols compared with CON. This result aligns with previous research suggesting that PerC

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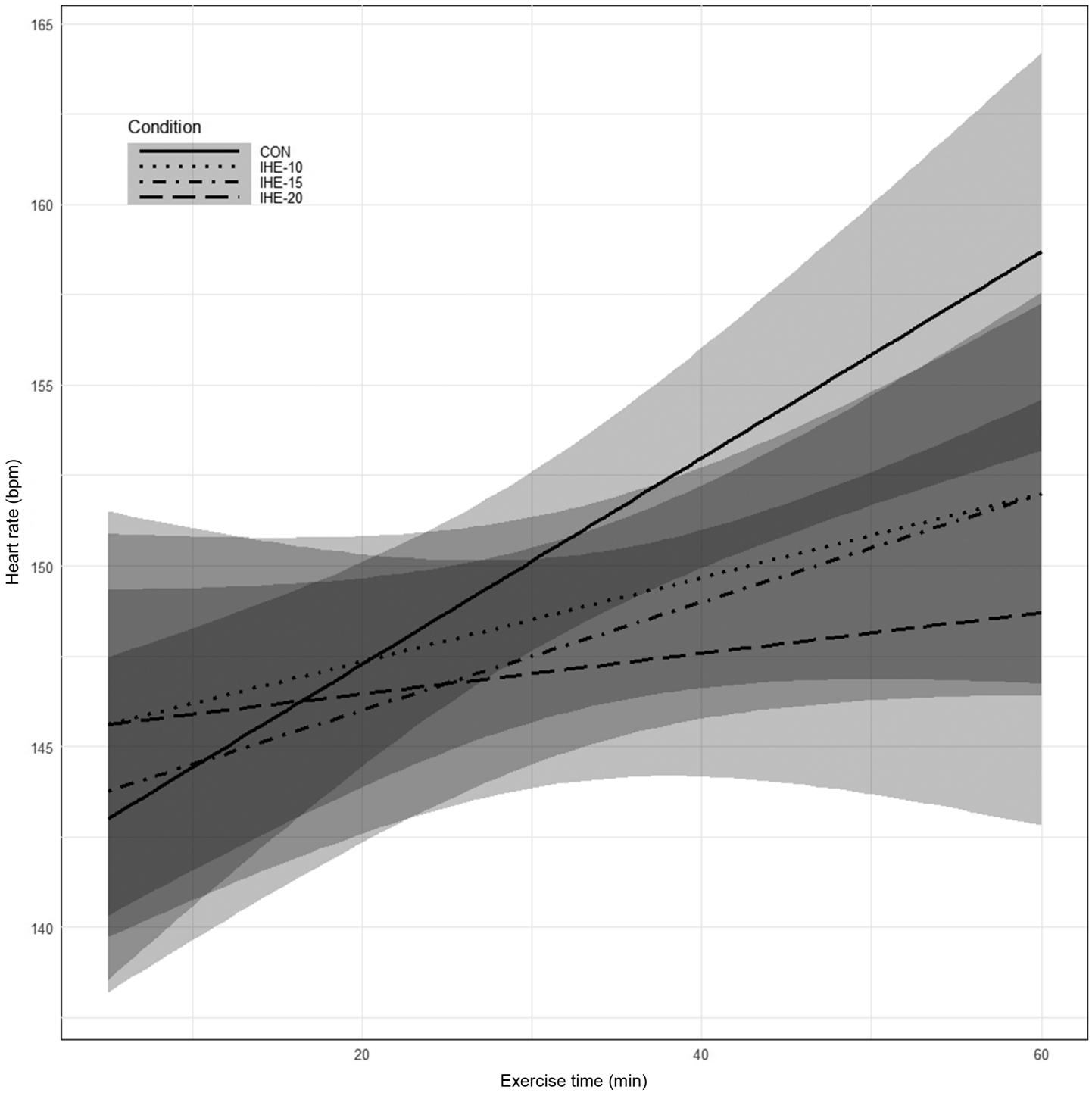

Figure 4 — Heart rate as a function of total time cycled in CON, 6 × 10-minute (IHE-10), 4 × 15-minute (IHE-15), and 3 × 20-minute (IHE-20) IHE conditions. Note: Shaded areas indicate 95% CIs. CON indicates control; IHE, intermittent heat exposure.

attenuates increases in T_c during exercise compared with scenarios without PerC and added breaks may further contribute to blunting T_c increases.[6,22,23] Additionally, a lower rate of increase was evident in IHE-20 compared with IHE-15 due to the longer rest break. However, no difference was observed in IHE-10 compared with IHE-15 and IHE-20. This indicates that IHE protocol selection may depend on the specific requirements of the training session, as different IHE set structures have varying effects on T_c.

To induce HA, it is commonly proposed that maintaining T_c of ≥ 38.5 °C is a key factor for adaptation.[24,25] Despite cooler T_b evident in all IHE protocols, IHE-10 spent ~48% of the total exercise time with T_c ≥ 38.5 °C, IHE-15 spent ~31%, and IHE-20 spent ~28%. Notably, no difference was observed between CON and IHE-10 for the percentage of total exercise time with T_c ≥ 38.5 °C (~50% vs ~48%). This may be due to greater total and mean 5-minute split distance completed in IHE-10 compared to

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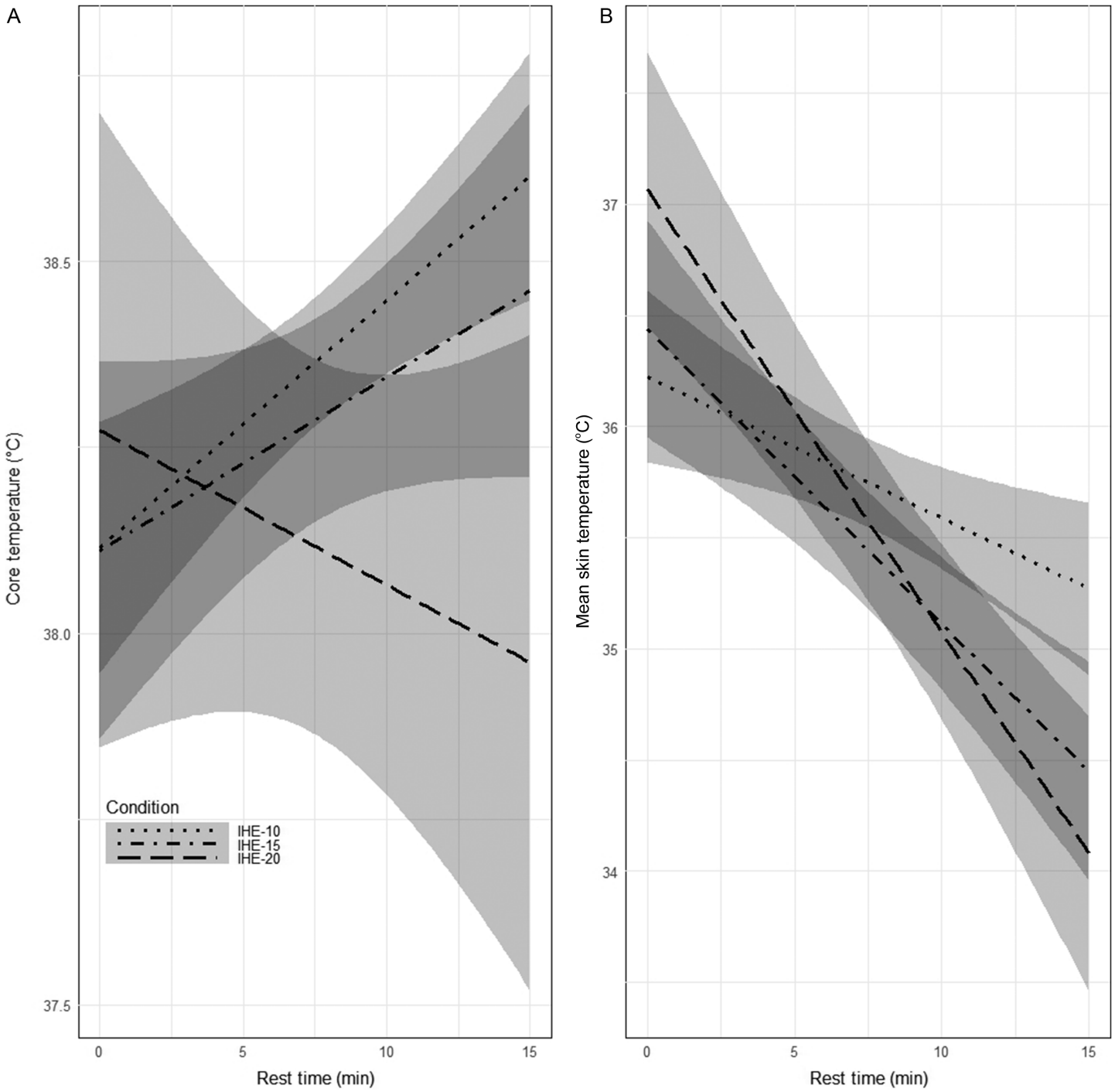

Figure 5 — (A) Core and (B) skin temperature as a function of total rest time in 6 × 10-minute (IHE-10), 4 × 15-minute (IHE-15), and 3 × 20-minute (IHE-20) IHE conditions. Note: Shaded areas indicate 95% confidence intervals. IHE indicates intermittent heat exposure.

CON, resulting in greater intensity-driven metabolic heat production equaling the continuous heat exposure thermal strain. Thus, an IHE protocol with shorter and more frequent breaks (eg, IHE-10) may enhance performance while promoting maintenance of high T_c closer to that of continuous exposure despite cooling rest periods.

Additionally, there was consistently lower mean T_{sk} observed over time during all IHE protocols compared with CON. This aligns with previous literature, as PerC is known to delay increases in T_{sk}, ultimately resulting in improved performance.[6,22,23] Schlader et al[26] suggested that high T_{sk} (>35 °C[1,27]) initiates an anticipatory response, leading to lower initial work rates during exercise in heat. Alternatively, lower T_{sk} corresponds with improved levels of Th_s and Th_c, resulting in improved distance completed during a 60-minute cycling time-trial. Our findings support this, as the lower T_{sk} observed in all IHE protocols corresponded with improved Th_c, Th_s, and RPE, along with greater total and 5-minute split distance completed.

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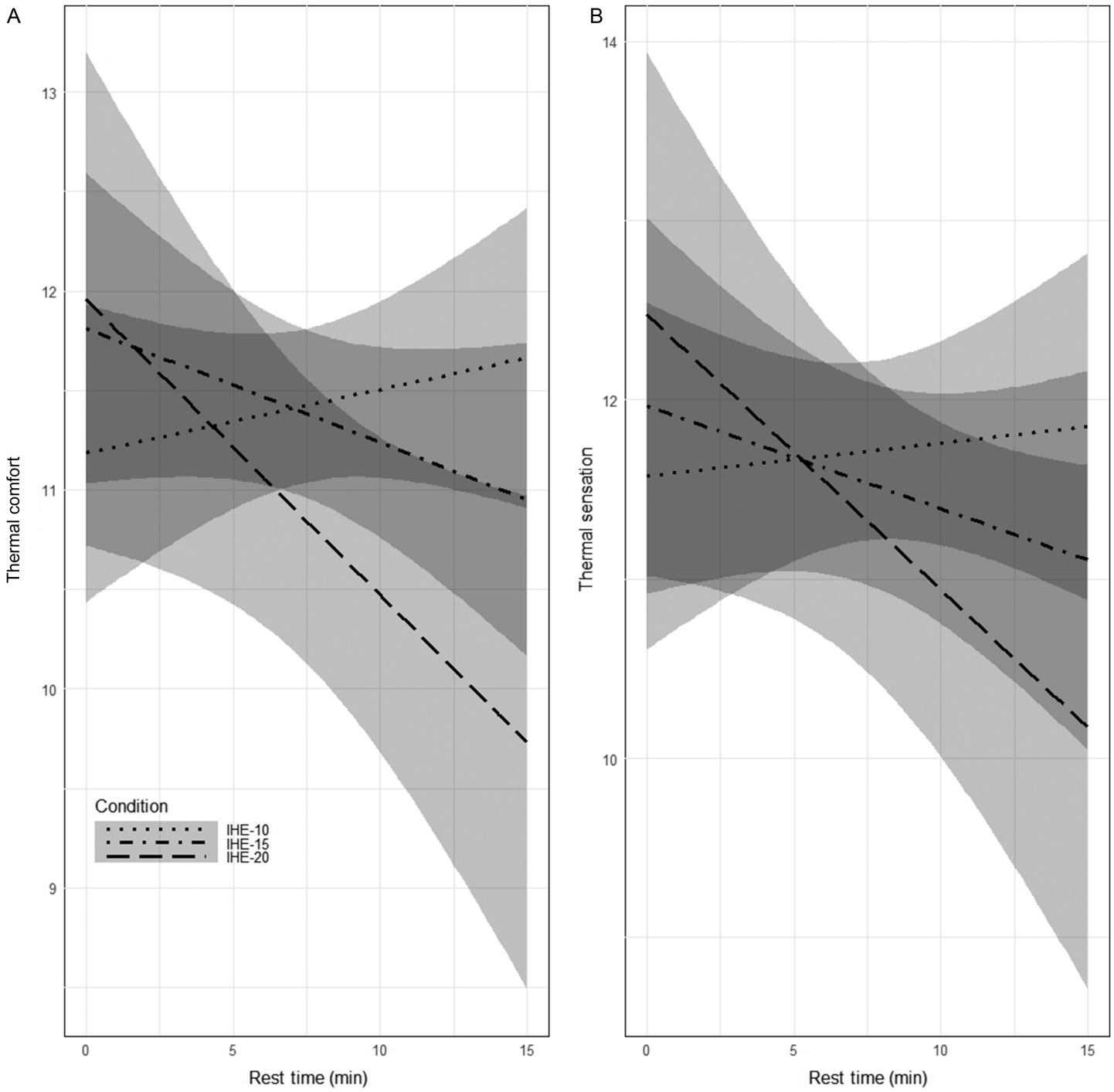

Figure 6 — (A) Thermal comfort and (B) thermal sensation as a function of total rest time in 6 × 10-minute (IHE-10), 4 × 15-minute (IHE-15), and 3 × 20-minute (IHE-20) IHE conditions. Note: Shaded areas indicate 95% CIs. IHE indicates intermittent heat exposure.

Sweat loss was not different between CON and IHE protocols, despite the intermittent nature of the exercise and the application of PerC, suggesting sweat rates are comparable between IHE and continuous heat exposure. Our study observed greater sweat loss in single IHE sessions (~2.05 L) compared with what Choo et al[7] and Naito et al[9] reported in their preacclimation sessions (1.15 and 1.32 L, respectively). This difference in sweat loss may be due to different levels of RH observed in both studies (~50% RH) compared with our study (~78% RH). This may have exacerbated the thermal strain experienced by participants in our study, as impaired sweat evaporation can limit heat loss due to low water vapor pressure differences between the environment and skin.[28]

Although findings of this study support the notion that IHE with PerC may be an alternative method of HA, it must be noted that our study only provides an acute assessment of this protocol. Thus, further research is required to determine the chronic effects of repeated exposures (ie, a HA program). Another limitation of the study lies in its execution on nonheat acclimatized individuals and

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indoor conditions. Therefore, it cannot be excluded that a greater/lesser magnitude of change in performance, physiological, and perceptual responses is observed with heat-acclimated individuals, or when performed outdoors with the addition of environmental factors (eg, radiant heat or wind). Consequently, the outcomes of this study may not be directly applicable to partially or fully heat-acclimated individuals. Additionally, athletes in this study were classified as levels 2 to 3 (trained or highly trained), and it remains uncertain whether athletes at levels 4 to 5 (elite or world class), individuals at level 0 (sedentary), or those who are not healthy would respond similarly. These results also do not provide insight into the effect of IHE alone, and it is possible that the intermittent rest breaks may account for the differences on physiological, perceptual, and performance variables, irrespective of the PerC intervention.

Practical Applications

Traditional continuous exercise heat exposure training is recommended when the overall training load permits prioritizing a maximal HA stimulus over training quality. Conversely, in situations where training quality takes precedence over HA stimulus or when training load needs to be limited, IHE-20 and IHE-15 protocols may be preferred. Finally, IHE-10 may be the preferred choice when both HA stimulus and training quality hold equal priority.

Conclusions

Intermittent heat exposure (IHE) with per-cooling (PerC) resulted in increased training intensity compared with traditional continuous heat exposure. Despite the shorter exercise duration and more frequent breaks in IHE-10, the necessary stimulus (eg, time above 38.5 °C core temperature) and physiological responses (eg, sweat loss) required for effective heat acclimation were not compromised. This can be attributed to lower core and skin temperature, as well as improved perceptual responses when PerC was applied.

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